

Bayesian Reformulation of *Suspecting Pharmaceuticals*

A probabilistic + decision-theoretic translation with simulations

Bayesian reformulation

This note reformulates the paper's central claims in a Bayesian framework where:

- The sponsor controls a **selection/framing channel** from full evidence to the observed dossier.
- Review is **two-phase**: *skeptical validity checks* followed by *suspicious integrity inference*.
- “Suspicious mood” is modeled as a **control state** that governs costly auditing and the probability of detecting diagnostic clues.
- Decisions implement **Bayesian decision theory** (“trustworthy enough” = a posterior expected-utility threshold).

Notation

- $\theta \in \Theta$: true causal effect (e.g., efficacy / safety composite).
- I : complete but unobserved evidence set (all trials/endpoints/analyses that exist).
- D : observed dossier (submitted/published/retrievable).
- $m \in \mathcal{M}$: manipulation/curation mechanism(s) (omission, framing, control substitution, wording inflation, ...).
- $v \in \{0, 1\}$: methodological validity (passes “standards” checks).
- $s \in [0, 1]$: suspicious mood intensity (epistemic control state).
- $a \in \mathcal{A}$: audit action (deep read, registry search, footnote chase, ...).
- X_a : audit-generated clues/features revealed under action a .

1. Full generative model

We model a world where the complete evidence I is generated from truth θ , but the observed dossier D is a **strategically filtered** function of I .

$$\theta \sim p(\theta).$$

$$I \sim p(I \mid \theta, \eta),$$

where η summarizes design, measurement noise, and heterogeneity.

$$m \sim p(m \mid \pi),$$

where π summarizes sponsor incentives and constraints.

$$D \sim p(D \mid I, m, \pi).$$

Because I is unobserved, the operative likelihood is:

$$p(D \mid \theta, m) = \int p(D \mid I, m, \pi) p(I \mid \theta, \eta) dI.$$

2. Joint inference over truth and integrity

The reviewer's joint posterior is:

$$p(\theta, m \mid D) \propto p(D \mid \theta, m) p(\theta) p(m).$$

Marginal effect inference is a mixture over integrity modes:

$$p(\theta \mid D) = \sum_m p(\theta \mid D, m) p(m \mid D).$$

3. Two-phase review: skepticism then suspicion

3.1 Skeptical phase: validity gating

$$p(v \mid D) \propto p(D \mid v) p(v).$$

Proceed to suspicious review only if the dossier “passes standards”:

$$p(v = 1 \mid D) \geq \tau_v.$$

3.2 Suspicious phase: integrity inference

conditional on validity

$$p(m \mid D, v = 1) \propto p(D \mid m, v = 1) p(m).$$

Effect inference (conditional on validity) is still a mixture:

$$p(\theta \mid D, v = 1) = \sum_m p(\theta \mid D, v = 1, m) p(m \mid D, v = 1).$$

4. Mooded labor as active information acquisition

Audit actions produce additional observations X_a ; mood s affects both **which** audits are chosen and **how** detectable clues are.

$$X_a \sim p(X_a \mid I, m, a, s).$$

Total evidence after auditing:

$$E = (D, X_a).$$

Updated posterior:

$$p(\theta, m \mid E) \propto p(E \mid \theta, m) p(\theta) p(m).$$

Audit policy (mood-conditioned):

$$\pi(a \mid D, s) = \Pr(a \text{ chosen} \mid D, s).$$

A simple detection model could be written as (higher suspicion improves diagnostic detection):

$$\Pr(\text{detect} \mid m = 1, a, s) = \sigma(\omega_0 + \omega_1 s + \omega_2 g(a)),$$

with logistic σ and audit-depth mapping $g(a)$.

5. Getting into the mood: dynamics and social “contamination”

Let $u \geq 0$ be effort invested to cultivate suspicion and r_t be social interaction/rapport (which empirically disrupts suspicion).

$$s_{t+1} = \rho s_t + \beta u_t - \gamma r_t + \epsilon_t.$$

A static saturation alternative:

$$s = 1 - \exp(-ku) + \epsilon.$$

6. “Trustworthy enough” as Bayesian decision theory

Let $y \in \{0, 1\}$ be approve/fund, and let θ_0 be a benefit threshold. Define asymmetric loss:

$$L(y, \theta) = \begin{cases} \lambda_{FP} & \text{if } y = 1 \text{ and } \theta < \theta_0 \\ \lambda_{FN} & \text{if } y = 0 \text{ and } \theta \geq \theta_0 \\ 0 & \text{otherwise.} \end{cases}$$

Optimal decision rule:

$$y^*(E) = \arg \min_y \mathbb{E}[L(y, \theta) \mid E].$$

Equivalent threshold form (for monotone utilities):

$$\text{Approve iff } \mathbb{E}[U(\theta) \mid E] \geq \kappa.$$

7. Value of information: the cost of suspicion

The reviewer chooses effort u and audit action a to maximize expected utility net of costs:

$$\max_{u, a} \mathbb{E}[-L(y^*(E), \theta) \mid D] - C(a, s) - G(u),$$

where s depends on u and context, $C(a, s)$ is the subjective/organizational cost (time, fatigue, social withdrawal). The ethnographic “closing doors / no lunch / avoid chatting” is the behavioral implementation of paying or managing C . And $G(u)$ is the effort cost of cultivating suspicion.